Basics of Statistics

The building blocks: data, variables, scales and frequency distributions.

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1 What is statistics?

Statistics is the science which deals with the **collection, organisation (classification), presentation, analysis and interpretation of data**.

- **Origin:** Italian *statista* (statesman), German *Statistik*, used by **Gottfried Achenwall** (1749).
- **Sir John Sinclair** brought the word into English through the *Statistical Account of Scotland* (1791–99, 21 volumes).
- **Sir R.A. Fisher** (1890–1962): **Father of Modern Statistics**. Gave us ANOVA, experimental design and maximum likelihood, and the principles of **randomisation, replication and local control**.

2 Two branches

Statistics

Descriptive	Inferential
Summarises a sample or population with measures such as the mean or standard deviation.	Uses a random sample to describe and draw inferences about population parameters.

Data are individual pieces of factual information: the raw material of statistics (e.g. rainfall over a period, area under paddy in a state).

3 Statistics vs maths

	Mathematics	Statistics
Reasoning	Deductive	Inductive
Results	Exact, unchanging	Under uncertainty
Handles	Axioms, proofs	Variable real-world data

The **error term** captures the variability we cannot control: observation − true model mean.

4 Population, sample, parameter

Population
The entire collection of units with the characteristic under study.
e.g. heights of **all** students in a college

Sample
A subset selected for study; should be **representative** of the population.
e.g. heights of 20 selected students

Parameter
A numerical characteristic of a population. It is **fixed** but usually **unknown**.
$\mu,\ \sigma^2,\ P$

Statistic
A value calculated from sample observations; used to **estimate** the parameter.
$\bar{x},\ s^2,\ \hat{p}$

5 Variables & constants

Variable: takes different values for different individuals, times or places.
Constant: stays fixed (e.g. $\pi = 3.14159\ldots$, $e = 2.71828\ldots$).

Type	Meaning	Agri examples
Discrete quantitative	Countable, separate values (0, 1, 2 …)	Fruits per branch, plots in a field
Continuous quantitative	Any value in a range	Plant height, yield, panicle length
Categorical qualitative	Categories, not measurements; can be counted	Variety, colour, disease present/absent

6 Measurement scales

Scale	Order	Equal gaps	True zero	Example
Nominal	×	×	×	Blood group, gender
Ordinal	✓	×	×	Grades (excellent, good …)
Interval	✓	✓	×	Temperature in °C
Ratio	✓	✓	✓	Weight, Kelvin

40 °C is **not** "twice as hot" as 20 °C (no true zero), but 60 kg **is** twice 30 kg.

7 Collection of data

Primary data
First-hand, collected for the study's own purpose; no prior statistical treatment.
e.g. population census by the government

Secondary data
Already collected and processed by others for another purpose.
e.g. Economic Survey, Ministry reports

PRIMARY METHODS personal investigation · trained enumerators · correspondents · phone or mailed questionnaire · online forms · mobile apps and social media polls

SECONDARY SOURCES official (RBI, ministries, NSO) · institutional (universities, research institutes). **Enumerator** collects; **respondent** provides.

8 Frequency distribution terms

Class limits	End points of a class, e.g. 110–119
Class boundaries	Close the gaps: 109.5–119.5 for data recorded to the nearest cm
Continuous classes	No gaps (10–20, 20–30): limits = boundaries; use $10 \le x < 20$
Class mark	Mid-point of the class interval
Class width	Upper boundary − lower boundary
LCF / GCF	Less-than (at upper boundaries) / greater-than (at lower boundaries) running totals
Relative freq.	f/N; all of them add up to 1

9 Building a grouped table

1. Decide the number of classes k
2. Determine the class width
3. Set up the class limits
4. Tally items into classes (every 5th stroke crosses the first four)
5. Count the frequency in each class

Raw data → **ungrouped** table (each value) or **grouped** table (classes). Grouping gives a clear picture but loses individual values.

10 Worked example · leaf length (cm), N = 38

Sturges: $k = 1 + 3.322 \log_{10} 38 \approx 6.25 \Rightarrow$ 6 classes; width $(18-1)/6 \approx 2.83 \Rightarrow 3$ cm.

Class	Boundaries	Mark	f	LCF	GCF	RF
1–3	0.5–3.5	2	3	3	38	0.079
4–6	3.5–6.5	5	6	9	35	0.158
7–9	6.5–9.5	8	10	19	29	0.263
10–12	9.5–12.5	11	5	24	19	0.132
13–15	12.5–15.5	14	5	29	14	0.132
16–18	15.5–18.5	17	9	38	9	0.237

11 Formulae to remember

Number of classes (Sturges)

$$k = 1 + 3.322 \log_{10} N$$

Approximate class width

$$C = \frac{|\max - \min|}{k}$$

Lower boundary of class i

$$L_i = \frac{\mathrm{LCL}_i + \mathrm{UCL}_{i-1}}{2}$$

Class mark

$$\frac{\text{Lower limit} + \text{Upper limit}}{2}$$

Relative frequency

$$\mathrm{RF} = \frac{f}{N}, \quad \sum \frac{f}{N} = 1$$

Sum of frequencies

$$\sum f = N$$

LCL / UCL = lower / upper class limit. Sturges' k is a guide, not a compulsory number.

! Common mistakes

- × A large dataset cannot correct biased selection or poor measurement.
- × Association alone does not prove cause and effect.
- × Online forms and social media polls collect voluntary responses, so they are not automatically representative.
- × Classes must cover the data **without overlap**. Boundaries depend on the recording precision.

? Fill in the blanks

1. Weight of a harvested mango recorded as 152.4 g is a _____ variable. **[continuous]**
2. The average yield of 100 sampled paddy farms is a _____; the true average of every farm in the district is a _____. **[statistic; parameter]**
3. Temperature in °C is on an _____ scale, so a 40 °C day is not "twice as hot" as a 20 °C day. **[interval]**
4. Rating farmers' satisfaction as low, medium or high uses the _____ scale. **[ordinal]**
5. A researcher using rainfall figures copied from an IMD report is working with _____ data. **[secondary]**
6. Plant height recorded to the nearest cm in the class 110–119 has class boundaries _____ and _____. **[109.5 and 119.5]**

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